

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

1. Missing OSI Layers

Concept Map:

Application → Presentation → Session → Transport → Network → Data Link → Physical

Missing Layers:

Session Layer

- Establishes, manages, and terminates communication sessions.
- Maintains synchronization between devices.

Data Link Layer

- Provides error detection and flow control.
- Uses MAC addresses for reliable local data delivery.

Conclusion:

These layers ensure reliable communication by managing connections and detecting transmission errors.

2. Concept Map for Data Units

Application / Presentation / Session

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Transport Layer → Segment

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Network Layer → Packet

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Data Link Layer → Frame

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Physical Layer → Bits

Explanation:

- Segment – Created by the Transport layer.
- Packet – Formed by adding the IP header.
- Frame – Formed by adding MAC header and trailer.
- Bits – Converted into electrical/optical/radio signals for transmission.

Conclusion:

Each layer adds its own information, ensuring accurate and reliable data transmission.

3. MAC Address and IP Address

Concept Map:

Data Link Layer → MAC Address → Local Delivery

Network Layer → IP Address → Global Delivery

Explanation:

- MAC Address identifies devices within the same local network (LAN).
- IP Address identifies devices across different networks (Internet).
- Together they enable end-to-end communication.

Conclusion:

MAC handles local delivery, while IP handles global routing.

4. Error Control

Concept Map:

Error Control

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Data Link Layer → Error Detection

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Transport Layer → Error Correction

Explanation:

- Data Link Layer detects errors using CRC or checksum.
- Transport Layer (TCP) retransmits lost or corrupted data and ensures correct delivery.

Conclusion:

The combination of error detection and correction improves communication reliability.

5. Concept Map for Web Browsing Using OSI Layers

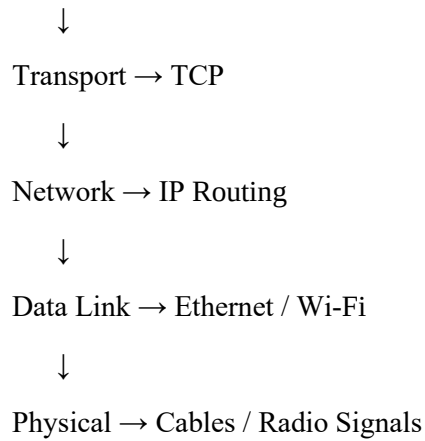
Application → HTTP/HTTPS

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Presentation → Encryption (SSL/TLS)

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Session → Session Management



Analysis:

- The Application layer sends the web request.
- The Presentation layer encrypts the data.
- The Session layer maintains the connection.
- TCP ensures reliable delivery.
- IP routes packets.
- Data Link delivers frames locally.
- Physical transmits bits.

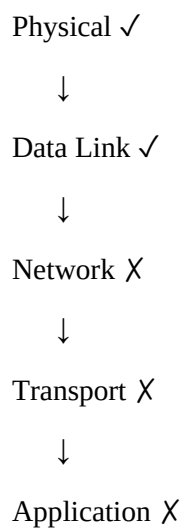
Conclusion:

Failure in any layer interrupts web browsing.

e.g., network or transport

6. Network Troubleshooting

Concept Map:



Analysis:

- Physical and Data Link layers are functioning.
- The Network layer has failed (possible IP configuration or routing issue).

- Since the Network layer fails, Transport and Application layers also fail.
- Troubleshooting Steps:
- Check IP address configuration.
- Verify subnet mask and default gateway.
- Test routing using ping or traceroute.
- Restart or reconfigure the router if required.

Conclusion:

The concept map helps identify that the problem starts at the Network layer, making troubleshooting faster and easier.